



Computer Networks 07 | Problem Solving on Subnetting part-3 | CS & IT | GATE Crash Course

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Summary

AI Notes

AI Slides

Text Notes

Screenshots

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Q.1

Consider three machines M, N, and P with IP addresses 157.157.38.90, 157.157.48.90, and 157.157.68.90 respectively. The subnet mask is set to 255.255.192.0 for all the three machines. Which one of the following is true?

A

M, N, and P belong to three different subnets

B

Only N and P belong to the same subnet

C

Only M and N belong to the same subnet

D

M, N, and P all belong to the same subnet

▶ 3:50

$$M = 157.157.00100110.90$$

$$\text{AND} \quad \text{AND}$$

$$\text{SM} = 255.255.11000000.0$$

$$\text{SID} = 157.157.00000000.0$$

$$\text{SID} = 157.157.0.0$$

$$N = 157.157.00110000.90$$

$$\text{AND} \quad \text{AND}$$

$$\text{SM} = 255.255.11000000.0$$

$$\text{SID} = 157.157.00000000.0$$

$$157.157.0.0$$

$$P = 157.157.01000100.90$$

$$\text{AND} \quad \text{AND}$$

$$\text{SM} = 255.255.11000000.0$$

$$\text{SID} = 157.157.01000000.0$$

$$157.157.64.0$$

▶ 12:18

Just Check the bits of IP address which is Set in S.M. It gives u SID.

▶ 12:18

Q.2 Consider three machines M, N, and P with IP addresses 157.157.38.90, 157.157.48.90, and 157.157.68.90 respectively. The subnet mask is set to 255.255.240.0 for all the three machines. Which one of the following is true?

- A M, N, and P belong to three different subnets
- B Only N and P belong to the same subnet
- C Only M and N belong to the same subnet
- D M, N, and P all belong to the same subnet

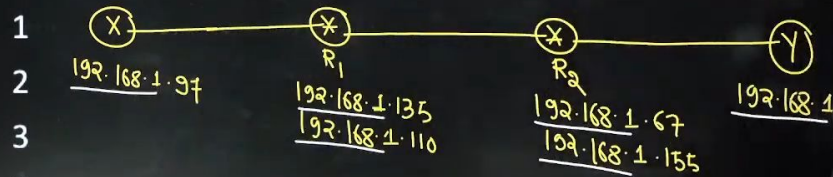
▶ 18:12

Q.5

Host X has IP address 192.168.1.97 and is connected through two routers R1 and R2 to another host Y with IP address 192.168.1.80. Router R1 has IP addresses 192.168.1.135 and 192.168.1.110. R2 has IP addresses 192.168.1.67 and 192.168.1.155. The netmask used in the network is 255.255.255.224.

Given the information above, how many distinct subnets are guaranteed to already exist in the network? **GATE IT 2006**

- A 1
- B 2
- C 3
- D 6



▶ 38:04

SM: 255.255.255.11100000

NID
SID HID

128 64 32

97: 0 1 1
135: 1 0 0
110: 0 1 1
67: 0 1 0
155: 1 0 0
80: 0 1 0

0 1 1
1 0 0
0 1 0 } 3 subnet

Q.6

The subnet mask for a particular network is 255.255.31.0. Which of the following pairs of IP addresses could belong to this network?



GATE CS 2003

- A 172.57.88.62 and 172.56.87.233
- B 10.35.28.2 and 10.35.29.4
- C 191.203.31.87 and 191.234.31.88
- D 128.8.129.43 and 128.8.161.55

▶ 38:07

good question

▶ 38:07

Q.7

Suppose computers A and B have IP addresses 10.105.1.113 and 10.105.1.91 respectively and they both use the same netmask N. Which of the values of N given below should not be used if A and B should belong to the same network? GATE CS 2010



255.255.255.0

255.255.255.128

255.255.255.192

255.255.255.224

A: 10.105.1.01110001
24
24
24
24

B: 10.105.1.01011011
24
24
24
24

a) 255.255.255.0

b) 255.255.255.10000000

c) 255.255.255.11000000

GATE WALLA

▶ 59:57

Q.8

The routing table of a router is shown below:

Destination	Subnet Mask	Interface
128.75.43.0	255.255.255.0	Eth0
128.75.43.0	255.255.255.128	Eth1
192.12.17.5	255.255.255.255	Eth3
default		Eth2

On which interfaces will be router forward packets addressed to destinations 128.75.43.16 **GATE 2004**

- ☐ A Eth1
 ☐ B Eth0
 ☐ C Eth3
 ☐ D Eth2

▶ 1:03:49

Q.8

The routing table of a router is shown below:

Destination (NID)	Subnet Mask	Interface
128.75.43.0 ✓	<u>255.255.255.0</u>	<u>Eth0</u> ✓✓
128.75.43.0	255.255.255.128	Eth1
192.12.17.5 ✓	255.255.255.255	Eth3
default		Eth2

On which interfaces will be router forward packet addressed to destinations 128.75.43.16 **GATE 200**

- ☐ A Eth1
 ☐ B Eth0
 ☐ C Eth3
 ☐ D Eth2

▶ 1:10:07

(Destination id) DID & SM = SID or NID

▶ 1:10:07

Note

If DIP matched with more than one interface then Router forward the Packet to the Interface which Have Longest subnet Mask [i.e more No. of 1's in the subnet mask]

▶ 1:15:59

Why longest subnet mask is preferred? (not imp)

▶ 1:17:08

Trick-> Start with the longest subnet to save time.

▶ 1:18:27

Q.9

A router uses the following routing table:

Destination (NID)	Mask	Interface
144.16.0.0	255.255.0.0 ✓	eth0
144.16.64.0	255.255.224.0 ✓	eth1
<u>144.16.68.0</u>	<u>255.255.255.0</u>	<u>eth2</u>
<u>144.16.68.64</u>	255.255.255.224 ✗	eth3

A packet bearing a destination address 144.16.68.117 arrives at the router. On which interface will it be forwarded?

(A) Eth0

(B) Eth1

(C) Eth2

(D) Eth3

▶ 1:25:22

PW

Note
First start with the longest subnet mask

I

DIP	144.16.68.01110101
AND	AND
SM	255.255.255.11100000
NID of SID	= 144.16.68.01100000
	= 144.16.68.96

Not matched with Eth3

II

DIP	= 144.16.68.117
AND	
SM	= 255.255.255.0
NID	= 144.16.68.0

Matched with Eth2

▶ 1:25:23

Q.10

An IP router implementing Classless Inter-domain Routing (CIDR) receives a packet with address 131.23.151.76. The router's routing table has the following entries:

Prefix	Output Interface Identifier
131.16.0.0/12	3
131.28.0.0/14	5
131.19.0.0/16	2
131.22.0.0/15	1

The identifier of the output interface on which this packet will be forwarded is ____.

GATE 2014

▶ 1:25:53

Q.11

A company has a class C network address of 204.204.204.0. It wishes to have three subnets (X,Y,Z) X with 100 hosts, Y with 50 hosts and Z with 50 host. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?



MSQ

- A** X-204.204.204.0/255.255.255.128
Y-204.204.204.128/255.255.255.192
Z-204.204.204.192/255.255.255.192
- C** X-204.204.204.128/255.255.255.128
Y-204.204.204.0/255.255.255.192
Z-204.204.204.64/255.255.255.192
- B** X-204.204.204.0/255.255.255.128
Y-204.204.204.192/255.255.255.192
Z-204.204.204.128/255.255.255.192
- D** X-204.204.204.128/255.255.255.128
Y-204.204.204.64/255.255.255.192
Z-204.204.204.0/255.255.255.192

▶ 1:27:22

204.204.204.0



Class-C

NID	HID
24	8

3 subnet

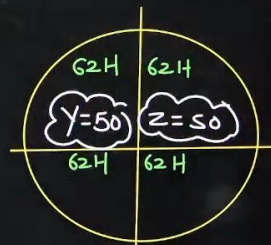
$\frac{2}{SID} \frac{6}{HID} \rightarrow 2^6 - 2 = 62 \text{ Host/subnet}$
 $2^2 = 4 \text{ subnet}$

X=100 Host

Y=50 Host

Z=50 Host

$200 \leq 2^8 - 2$ (Yes)



Not Possible

▶ 1:34:02

Diagram of a 28-bit IP address structure. The address is divided into four 7-bit IP segments. The first segment is labeled 'X=100H'. The second segment is labeled 'Y=50H'. The third segment is labeled 'Z=50H'. The fourth segment is labeled 'SID'. The first three segments are each labeled '7 bit IP'. The fourth segment is labeled 'SID' and '6 bit HIP'.

Y

SID: 204.204.204.128
SM: 255.255.255.192 (128+64)
No. of 1's in SM = NID + SID = 24 + 2 = 26
||||||| . ||||| . ||||| . ||000000 + 255.255.255.192

Z

SID: 204.204.204.192
SM: 255.255.255.192 (128+64)
No. of 1's in the SM = NID + SID = 24 + 2 = 26
IInd way
X
Y
Z

X

SID: 204.204.204.0
SM: 255.255.255.128
No. of 1's in the SM = NID + SID = 24 + 1 = 25
No. of 0's in the SM = NID = 7
||||||| . ||||| . ||||| . |0000000
255.255.255.128

▶ 1:51:24

OR

Diagram of a 28-bit IP address structure. The address is divided into four 7-bit IP segments. The first segment is labeled 'X=100H'. The second segment is labeled 'Y=50H'. The third segment is labeled 'Z=50H'. The fourth segment is labeled 'SID'. The first three segments are each labeled '7 bit IP'. The fourth segment is labeled 'SID' and '6 bit HIP'.

Y

SID: 204.204.204.0
SM: 255.255.255.192 (128+64)

Z

SID: 204.204.204.64
SM: 255.255.255.192 (128+64)

X
Y
Z

X

SID: 204.204.204.128
SM: 255.255.255.128

▶ 1:58:52

Q.12

The address of a class B host is to be split into subnets with a 6-bit subnet number. What is the maximum number of subnets and the maximum number of hosts in each subnet?
GATE 2007

62 subnets and 262142 hosts.

64 subnets and 262142 hosts.

62 subnets and 1022 hosts.

64 subnets and 1024 hosts.

64 subnets and 1022 Host

Class-B

$\frac{NID}{16} \quad \frac{HID}{16}$

$\frac{6}{SID} \quad \frac{10}{HID}$
 $2^6 = 64 \text{ subnets}$
 $2^{10} - 2 = 1022 \text{ Host/sub}$

▶ 2:05:23

RFC 950

$SID = n \text{ bit}$

No. of subnets = $2^n - 2$

$SID = 3 \text{ bit}$

$S_1 - 000 \times$
 $S_2 - 001$
 $S_3 - 010$
 $S_4 - 011$
 $S_5 - 100$
 $S_6 - 101$
 $S_7 - 110$
 $S_8 - 111 \times$

Latest RFC

1812

$SID = n \text{ bit}$

No. of subnets = 2^n

▶ 2:05:39

Q.13

Using the IP address 172.168.42.58 and subnet mask 255.255.252.0, identify the correct subnet ID and directed broadcast address.

(Assuming Classfull addressing scheme is followed).

The correct Network ID is 172.168.40.0, and the broadcast address is 172.168.255.255

The correct Network ID is 172.168.40.0, and the broadcast address is 172.168.43.255

The correct Network ID is 172.168.40.0, and the broadcast address is 172.168.44.255

The correct Network ID is 172.168.40.0, and the broadcast address is 172.169.43.255

subnet

▶ 2:05:48

Q.14

Consider the following IP address 200.48.67.184 and subnet mask 255.255.255.240, what is the IP address of last host of subnet to which given IP address belongs?

- A 200.48.67.192
- B 200.48.67.190
- C 200.48.67.255
- D 200.48.67.254

▶ 2:06:23

Q.15

In a class B subnet, we know the IP address of one host and the mask as given below:

IP address: 125.134.112.66

Mask: 255.255.224.0

What is the first address (Network address)?

- A 125.134.96.0
- B 125.134.112.0
- C 125.134.112.66
- D 125.134.0.0

▶ 2:06:33

Q.16

Which of following IPs may belong to last host of any subnet if subnet mask is 255.255.255.224.

- (i) 210.15.16.62
- (ii) 210.15.16.94
- (iii) 210.15.16.127
- (iv) 210.15.16.191

- A I and ii
- B I and iii
- C li and iv

lii and iv

▶ 2:06:40

Q.17

Given the following: **MSQ**

400



Host IP address: 192.168.100.66, with 3 bits of subnetting.

Which of the following is/are true for the above network and host?

- A The subnet address to which this host belongs is 192.168.100.32
- B The subnet address to which this host belongs is 192.168.100.64
- C Broadcast address is 192.168.100.255
- D Valid host range is 192.168.100.65 to 192.168.100.127

▶ 2:06:49